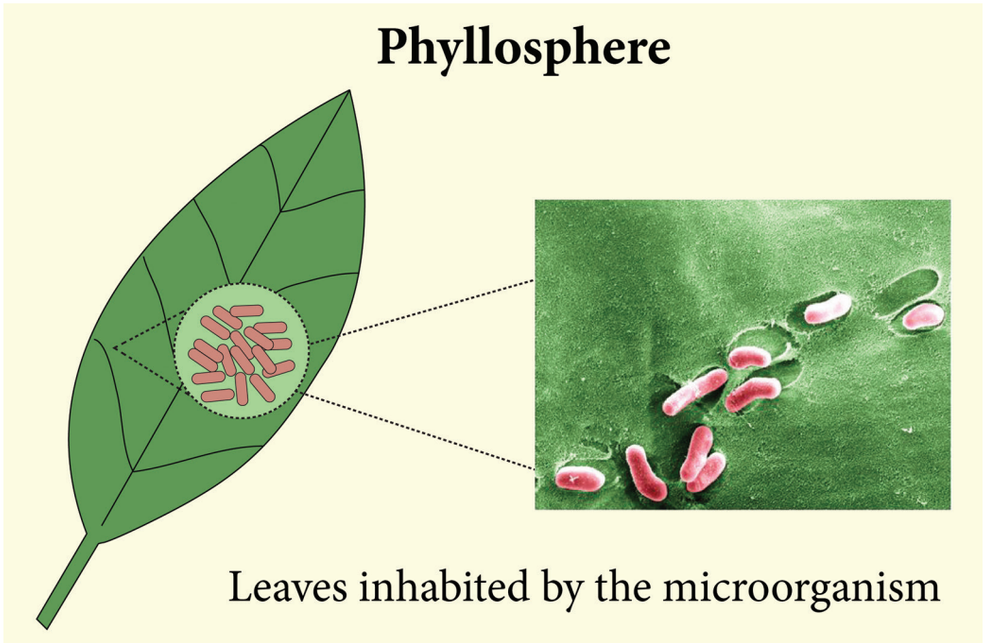


Figure 10.7: Microscopic appearance of Phyllosphere Bacteria

Infobits

PGPB can promote plant growth. The bacteria include those that are free-living, those that form specific symbiotic relationships with plants (Example: *Rhizobia* and *Frankia*), bacterial endophytes that can colonize some or a portion of a plant's interior tissues, and cyanobacteria (blue-green algae). PGPB may promote plant growth directly usually by either facilitating resource acquisition or modulating plant hormone levels, or indirectly by decreasing the inhibitory effects of various pathogenic agents on plant growth and development, that is, by acting as biocontrol bacteria. It is envisioned that in the not too distant future, plant growth-promoting bacteria (PGPB) will begin to replace the use of chemicals in Agriculture, Horticulture, Silviculture, and environmental cleanup strategies.

associations exist on leaves. The dominant and useful microorganisms on the leaf surfaces in the forest, vegetation happened to be Nitrogen fixing bacteria like *Beijerinckia* and *Azotobacter*.

Apart from Nitrogen fixing bacteria, other genera such as *Pseudomonas, Pseudobacterium, Phytomonas* are also encountered on the leaf surface. The age of plant, its leaf spread, morphology and maturity level and the atmospheric factors greatly influence the phyllosphere microflora.

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